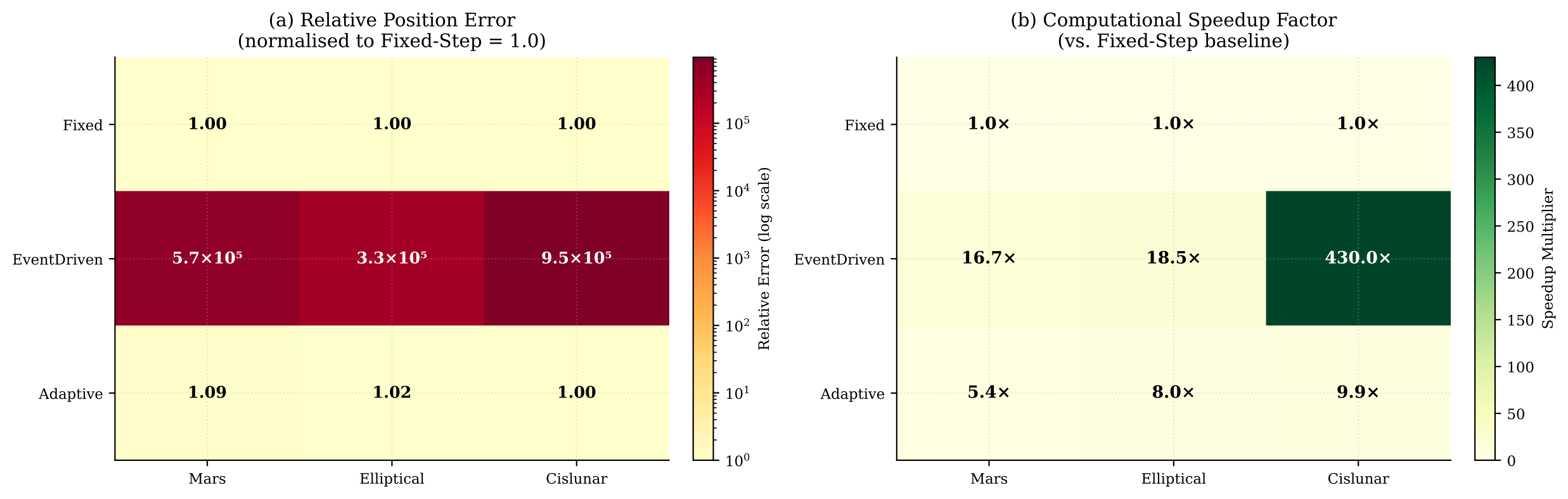
(a) Relative Position Error
(normalised to Fixed-Step = 1.0)
Fixed
EventDriven
Adaptive
1.00
1.00
1.00
5.7×10⁵
3.3×10⁵
9.5×10⁵
1.09
1.02
1.00
Mars
Elliptical
Cislunar
Relative Error (log scale)
10⁵
10⁴
10³
10²
10¹
10⁰
(b) Computational Speedup Factor
(vs. Fixed-Step baseline)
1.0×
1.0×
1.0×
16.7×
18.5×
430.0×
5.4×
8.0×
9.9×
Speedup Multiplier
400
350
300
250
200
150
100
50
0